PROJECT 1 OF 2 · SATELLITE REMOTE SENSING + DEEP LEARNING

SWOT Tank Water Tracker

How much water does each tank hold right now?

AI joins water **area**, mapped from Sentinel-1 and 2, with water **height** from SWOT (NASA and CNES, 2022) to give storage for every tank in a dry-zone cascade.

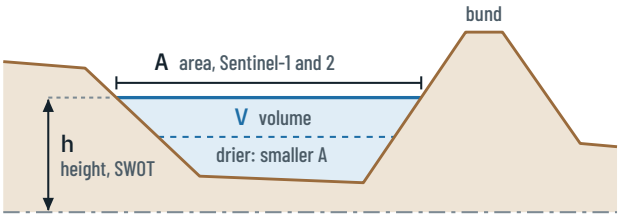
STUDY AREA	One dry-zone cascade + 2–3 major reservoirs; candidate: Anuradhapura
USERS	Irrigation staff, farmer organisations
COST	Free data and software; travel only
PLAN	Oct 2026 – Jun 2027 · 3 members

THE PROBLEM

Satellites already show where tank water is, but not how much each tank holds.

- **Area is not volume.** IFAD and ESA (2026) map 10,650+ water bodies in six districts monthly at 10 m, area only. A shallow and a deep tank of equal area hold very different amounts.
- **Small tanks go unmeasured.** About 23,000 tanks of 80 ha or less, about 21% non-functional; storage is reported mainly for major reservoirs.
- **Clouds and plants hide water.** Monsoon cloud blocks optical images; water hyacinth, lotus and Salvinia infest many seasonal tanks in Anuradhapura.
- **Height was out of reach.** Older altimeters saw only a narrow strip under their path; SWOT's wide swath is specified for lakes down to 250 m × 250 m.

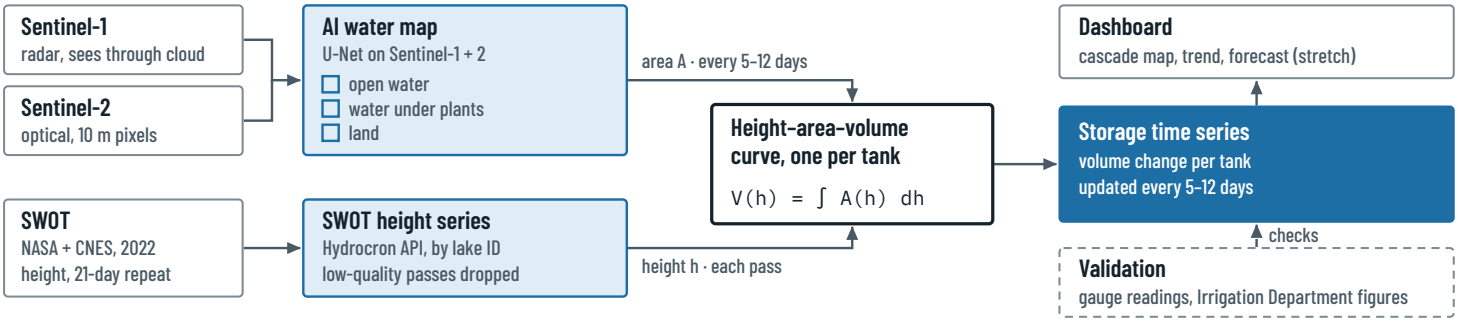
THE KEY IDEA



Each SWOT pass gives height *h* on a date when Sentinel also maps area *A*. A curve fitted per tank turns height into volume change; a capacity table, where one exists, gives % full.

HOW IT WORKS

two satellite inputs, one storage output



SWOT data since July 2023 fits each tank's curve; then every new Sentinel map becomes storage, every 5–12 days instead of 21.

WHAT IS NEW

● covered ○ not

SWOT storage studies exist abroad; we found none using AI water maps for Sri Lankan tanks.

STUDY	SRI LANKA	SWOT HEIGHT	VOLUME	SMALL LAKES
Ma et al. 2026 USA	○	●	●	○
Das & Hossain 2025	○	●	●	○
Wang & Tseng 2025 Taiwan	○	●	○	●
IFAD & ESA 2026	●	○	○	●
UoM studies 2018, 2024	●	○	○	○
This project	●	●	●	●

HOW WE PROVE IT

TESTED	METRIC	AGAINST
Water map	IoU, F1; cloud and plant cases reported apart	Water-index thresholds, IFAD/ESA maps
SWOT height	Error in cm against gauges, by tank size	Spec: 10 cm above 1 km²; 25 cm from 250 × 250 m to 1 km²
Storage	Error, % of capacity	Fixed-depth estimate
Forecast	Error 2 and 4 weeks ahead (stretch)	"No change" baseline

RESEARCH QUESTIONS AND OWNERS

- A RQ1 How accurately can deep learning map tank water from Sentinel-1 and 2, even under cloud or floating plants?
- B RQ2 How accurate is SWOT water height on tanks of different sizes, against gauge readings?
- B RQ3 Does AI water area plus SWOT height give storage close to official and measured values?
- C RQ4 Can storage be forecast 2–4 weeks ahead from recent storage and rainfall? STRETCH

A water mapping **B** SWOT, storage curves, validation
C pipeline, forecast, dashboard

SCOPE

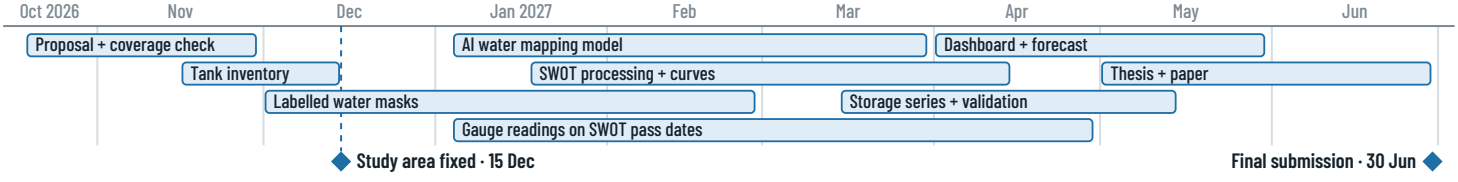
CORE	Water map with plant class, SWOT heights, storage curves, validation, dashboard
STRETCH	Forecast, SWOT pixel cloud, cloud removal, second cascade
OUT	Whole island, mobile app, real-time alerts, bed surveys

MAIN RISKS

- ▲ **Small tanks may lack SWOT lake IDs.** Count them before 15 Dec.
- ▲ **Heavy rain degrades Ka-band data.** Drop poor-quality passes.
- ▲ **Slow gauge records.** Ask in month one for records since 2023.

PLAN

proposed; dates assumed, adjust to faculty deadlines



SOURCES IFAD 2026 · Lopane et al. 2025 · Ma et al. 2026 · Wang & Tseng 2025 · Das & Hossain 2025 · NASA JPL 2022 · NASA ARSET 2025 · SWOT Science Team 2024 · AVISO · Hydrocron · IWMI · Madhurshan et al. 2024 · Meyen 2018

8.31°N 80.40°E · Anuradhapura